

SONNET PODDER

PERSONAL DATA

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PROFILE

I am currently pursuing an MSc in Big Data and Data Science Technology at Northumbria University, UK. With a strong background in Statistics and Data Science, my research interests include machine learning, statistical learning, optimization, explainable AI, and big data analytics. I am seeking PhD opportunities to develop interpretable, scalable AI and data-driven solutions for real-world challenges across interdisciplinary domains.

RESEARCH INTERESTS

Machine Learning, Deep Learning, Statistical Learning, Explainable AI, Predictive Modelling, Bayesian Statistics, Statistical Inference, Probability Theory, Uncertainty Quantification, Big Data Analytics, Data Mining, Statistical Computing, Optimization Techniques, Federated and Distributed Learning, with applications in Public Health Analytics, Bioinformatics, and Transportation Safety Analytics.

EDUCATION

Northumbria University London, UK MSc Big Data and Data Science Technology (with Advanced Practice) Result: Distinction Level (Current Average: 70.5%)	January 2026 – January 2028.
Gopalganj Science and Technology University, Bangladesh. Master of Science Statistics CGPA: 3.46 out of 4	July 2022 – June 2023.
Gopalganj Science and Technology University, Bangladesh. Bachelor of Science Statistics CGPA: 3.23 out of 4	January 2017 – June 2022.

ACADEMIC PROJECTS AND RESEARCH

- Credit Card Fraud Detection Using Machine Learning** - Developed machine learning models for highly imbalanced financial transaction data.

Used Methods - Logistic Regression, Random Forest, XGBoost, Neural Network, SMOTE, Random Under sampling, Cost-sensitive Learning, Stratified Cross Validation,

Hyperparameter Tuning.

2. **Big Data Analytics of Drug-Related Deaths** - Performed large-scale Python-based analysis of accidental drug-related deaths (2012–2024).

Used Methods - Data Cleaning, EDA, Statistical Testing, ANOVA, Correlation Analysis, Visualization

3. **Machine Learning on Cloud: Technical Evaluation and Reflection** - Critically evaluated the fraud detection ML project from a technical and professional perspective, covering preprocessing decisions, model selection rationale and responsible AI deployment considerations.

Used Methods - Duplicate Removal, Log Transformation, Feature Engineering, Standard Scaling, Model Comparison, False Positive/Negative Analysis, Interpretability Assessment, Fairness Evaluation, Model Monitoring.

4. **Business Intelligence Dashboard Using Tableau** - Developed interactive dashboards from Olympic Games data to explore medal patterns across city, season, athlete age and gender.

Used Tools - Tableau, Interactive Filters, Connected Charts, Dynamic Visualisation, Dashboard Design, Visual Layout Structuring.

5. **Machine Learning-Based Identification of Risk Factors and Determinants of Depressive Symptoms Among Elderly Bangladeshi Population** (*under preparation*) - Mohammad Kamal Hossain, Sonnet Podder. Developed a predictive modelling framework to analyse socio-demographic and health-related risk factors of geriatric depression among the elderly Bangladeshi population.

6. **Thesis Project: Statistical Analysis of Meteorological Influence on COVID-19 Transmission in Bangladesh** - Investigated the association between COVID-19 confirmed cases and meteorological indicators including temperature and rainfall across regional epidemiological datasets in Bangladesh.

Used Methods - Spearman Correlation, Kendall Correlation, Quantile Regression, OLS Regression, Non-parametric Statistical Analysis, Python, R, SPSS.

SKILLS AND COMPETENCES

Programming Knowledge: Python (NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow, Matplotlib, Seaborn), R, SQL.

Machine Learning: Supervised Learning, Ensemble Methods (Random Forest, XGBoost), Neural Networks, Imbalanced Learning (SMOTE, Cost-sensitive Learning), Stratified Cross-Validation, Hyperparameter Tuning, PR-AUC / ROC-AUC Evaluation

Statistical Methods: Regression Modelling, Quantile Regression, Non-parametric Methods, Statistical Learning, Bayesian Statistics, Hypothesis

	Testing, Inferential Statistics, Probability Theory
Software Knowledge:	MATLAB, SPSS, STATA, JMP, Minitab, Power BI, Tableau, EndNote, Zotero.
Cloud & Tools:	Google Colab, AWS (Cloud ML Workflows), Jupyter Notebook, GitHub
Data Analysis:	Data Cleaning, Exploratory Data Analysis, Feature Engineering, Statistical Modelling, Data Visualisation, Large-scale Data Analysis.
Research Skills:	Statistical modeling, Machine learning, predictive analytics, uncertainty quantification, risk analysis, data preprocessing, feature engineering, and large-scale data analysis using Python and R.

LANGUAGES PROFICIENCY

BENGALI: Mother Tongue

ENGLISH: Competent User [IELTS- 6.5 (L- 6.5, W-7, R- 6, S-6)]